



Multiple Random Variables Solutions

The concept of multiple random variables (bivariate random variables) deals with the **joint probability** of two variables, X and Y , occurring simultaneously.

1. Finding the Constant k in a Joint PMF

Question: The joint density function (Joint PMF) of X and Y is given by $P(x, y) = k(2x + y)$; $x = 0, 1, 2$; $y = 0, 1, 2$. Find the value of k .

Solution:

For any joint probability distribution (discrete or continuous), the sum (or integral) of all probabilities must equal 1. For a discrete joint PMF, we use the property:

$$\sum_x \sum_y P(x, y) = 1$$

1. **Calculate $P(x, y)$ for all pairs:**

- $x = 0$: $P(0, y) = k(0 + y)$
 - $P(0, 0) = 0k$
 - $P(0, 1) = 1k$
 - $P(0, 2) = 2k$
- $x = 1$: $P(1, y) = k(2 + y)$
 - $P(1, 0) = 2k$
 - $P(1, 1) = 3k$
 - $P(1, 2) = 4k$
- $x = 2$: $P(2, y) = k(4 + y)$
 - $P(2, 0) = 4k$
 - $P(2, 1) = 5k$
 - $P(2, 2) = 6k$

2. **Sum all probabilities and set equal to 1:**

$$0k + 1k + 2k + 2k + 3k + 4k + 4k + 5k + 6k = 1$$

$$(0 + 1 + 2 + 2 + 3 + 4 + 4 + 5 + 6)k = 1$$

$$27k = 1$$

3. Solve for k :

$$k = \frac{1}{27}$$

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2. Finding Marginal Distributions

Question: For the joint probability distribution of two random variables X and Y given below, find the **marginal probability distribution** of X and Y .

$X \setminus Y$	1	2	3	4	$P(X)$ (Row Total)
1	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{10}{36}$
2	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{9}{36}$
3	$\frac{5}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{8}{36}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{5}{36}$	$\frac{9}{36}$
$P(Y)$ (Col. Total)	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	1

Solution:

The **marginal probability distribution** for a variable is found by summing the joint probabilities across all values of the other variable.

1. **Marginal Distribution of X ($P(X)$):** (Summing the rows)

- $P(X = 1) = \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{10}{36}$
- $P(X = 2) = \frac{1}{36} + \frac{3}{36} + \frac{3}{36} + \frac{2}{36} = \frac{9}{36}$
- $P(X = 3) = \frac{5}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} = \frac{8}{36}$
- $P(X = 4) = \frac{1}{36} + \frac{2}{36} + \frac{1}{36} + \frac{5}{36} = \frac{9}{36}$

Marginal Distribution of X :

X	1	2	3	4
$P(X)$	$\frac{10}{36}$	$\frac{9}{36}$	$\frac{8}{36}$	$\frac{9}{36}$

2. Marginal Distribution of Y ($P(Y)$): (Summing the columns)

- $P(Y = 1) = \frac{4}{36} + \frac{1}{36} + \frac{5}{36} + \frac{1}{36} = \frac{11}{36}$
- $P(Y = 2) = \frac{3}{36} + \frac{3}{36} + \frac{1}{36} + \frac{2}{36} = \frac{9}{36}$
- $P(Y = 3) = \frac{2}{36} + \frac{3}{36} + \frac{1}{36} + \frac{1}{36} = \frac{7}{36}$
- $P(Y = 4) = \frac{1}{36} + \frac{2}{36} + \frac{1}{36} + \frac{5}{36} = \frac{9}{36}$

Marginal Distribution of Y :

Y	1	2	3	4
$P(Y)$	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$



3. Finding Conditional Probability Distribution

Question: The joint density function (Joint PMF) of X and Y is given by $P(x, y) = \frac{(2x+y)}{27}$; $x = 0, 1, 2$; $y = 0, 1, 2$. Find the **conditional probability distribution of Y given $X = x$** .

Solution:

The formula for the conditional PMF of Y given a specific value of X is:

$$P(Y|X = x) = \frac{P(x, y)}{P(x)}$$

where $P(x)$ is the marginal probability of X .

1. Find the Marginal Probabilities of X ($P(x)$):

We need the sums of the joint probabilities for each column x :

- $P(X = 0)$: $P(0, 0) + P(0, 1) + P(0, 2)$

$$P(0) = \frac{(0)}{27} + \frac{(0+1)}{27} + \frac{(0+2)}{27} = \frac{0+1+2}{27} = \frac{3}{27}$$

- $P(X = 1)$: $P(1, 0) + P(1, 1) + P(1, 2)$

$$P(1) = \frac{(2+0)}{27} + \frac{(2+1)}{27} + \frac{(2+2)}{27} = \frac{2+3+4}{27} = \frac{9}{27}$$

- $P(X = 2)$: $P(2, 0) + P(2, 1) + P(2, 2)$

$$P(2) = \frac{(4+0)}{27} + \frac{(4+1)}{27} + \frac{(4+2)}{27} = \frac{4+5+6}{27} = \frac{15}{27}$$

2. Calculate Conditional Probabilities for each fixed x :

- **Case 1:** $X = 0$ ($P(0) = 3/27$)

$$P(Y = y|X = 0) = \frac{P(0, y)}{P(0)} = \frac{(2(0) + y)/27}{3/27} = \frac{y}{3}$$

- $P(Y = 0|X = 0) = 0/3 = 0$
- $P(Y = 1|X = 0) = 1/3 = \frac{1}{3}$
- $P(Y = 2|X = 0) = 2/3 = \frac{2}{3}$

- **Case 2:** $X = 1$ ($P(1) = 9/27$)

$$P(Y = y|X = 1) = \frac{P(1, y)}{P(1)} = \frac{(2(1) + y)/27}{9/27} = \frac{2 + y}{9}$$

- $P(Y = 0|X = 1) = 2/9 = \frac{2}{9}$
- $P(Y = 1|X = 1) = 3/9 = \frac{3}{9}$
- $P(Y = 2|X = 1) = 4/9 = \frac{4}{9}$

- **Case 3:** $X = 2$ ($P(2) = 15/27$)

$$P(Y = y|X = 2) = \frac{P(2, y)}{P(2)} = \frac{(2(2) + y)/27}{15/27} = \frac{4 + y}{15}$$

- $P(Y = 0|X = 2) = 4/15 = \frac{4}{15}$
- $P(Y = 1|X = 2) = 5/15 = \frac{5}{15}$

- $P(Y = 2|X = 2) = 6/15 = \frac{6}{15}$



4. Marginal and Conditional Probability

Question: The joint probability distribution of two random variables X and Y is given by: $P(X = 0, Y = 1) = \frac{1}{3}$, $P(X = 1, Y = 1) = \frac{1}{3}$, and $P(X = 1, Y = -1) = \frac{1}{3}$. Find (i) Marginal Probability distribution of X and Y and (ii) Conditional Probability distribution of X given $Y = 1$.

Solution:

1. Construct the Joint PMF table:

The random variable X takes values 0 and 1. The random variable Y takes values -1 and 1.

$X \setminus Y$	-1	1	$P(X)$
0	0	$\frac{1}{3}$	$\frac{1}{3}$
1	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{3}$
$P(Y)$	$\frac{1}{3}$	$\frac{2}{3}$	1

i. Marginal Probability Distributions:

- **Marginal Distribution of X ($P(X)$):**

$$P(X = 0) = \frac{1}{3}$$

$$P(X = 1) = \frac{2}{3}$$

- **Marginal Distribution of Y ($P(Y)$):**

$$P(Y = -1) = \frac{1}{3}$$

$$P(Y = 1) = \frac{2}{3}$$

ii. Conditional Probability distribution of X given $Y = 1$:

The formula is $P(X = x|Y = 1) = \frac{P(x,1)}{P(Y=1)}$. We use the marginal probability $P(Y = 1) = 2/3$.

- $P(X = 0|Y = 1)$:

$$P(X = 0|Y = 1) = \frac{P(0,1)}{P(Y = 1)} = \frac{1/3}{2/3} = \frac{1}{2}$$

- $P(X = 1|Y = 1)$:

$$P(X = 1|Y = 1) = \frac{P(1,1)}{P(Y = 1)} = \frac{1/3}{2/3} = \frac{1}{2}$$

- **Conditional Distribution $P(X|Y = 1)$:**

X	0	1
$P(X Y = 1)$	$\frac{1}{2}$	$\frac{1}{2}$

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5. Joint PDF, Marginal, and Conditional PDF

Question: The joint probability density function (Joint PDF) of the two-dimensional random variable (X, Y) is of the form:

$$f(x, y) = \begin{cases} e^{-k(x+y)}, & 0 \leq y < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

- Determine the value of k .
- The marginal Probability density function.
- The conditional Probability density functions of X and Y .

Solution:

i. Determine the value of k

Use the total probability property for a Joint PDF:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

We integrate over the region $0 \leq y < x < \infty$:

$$\int_0^{\infty} \left(\int_0^x e^{-k(x+y)} dy \right) dx = 1$$

$$\int_0^{\infty} e^{-kx} \left(\int_0^x e^{-ky} dy \right) dx = 1$$

$$\int_0^{\infty} e^{-kx} \left[\frac{e^{-ky}}{-k} \right]_{y=0}^{y=x} dx = 1$$

$$\int_0^{\infty} e^{-kx} \left(\frac{e^{-kx}}{-k} - \frac{e^0}{-k} \right) dx = 1$$

$$\int_0^{\infty} e^{-kx} \left(\frac{1 - e^{-kx}}{k} \right) dx = 1$$

$$\frac{1}{k} \int_0^{\infty} (e^{-kx} - e^{-2kx}) dx = 1$$

$$\frac{1}{k} \left[\frac{e^{-kx}}{-k} - \frac{e^{-2kx}}{-2k} \right]_0^{\infty} = 1$$

$$\frac{1}{k} \left[(0 - 0) - \left(\frac{1}{-k} - \frac{1}{-2k} \right) \right] = 1$$

$$\frac{1}{k} \left[\frac{1}{k} - \frac{1}{2k} \right] = 1$$

$$\frac{1}{k} \left[\frac{2 - 1}{2k} \right] = 1$$

$$\frac{1}{k} \left(\frac{1}{2k} \right) = 1 \implies \frac{1}{2k^2} = 1$$

$$2k^2 = 1 \implies k^2 = \frac{1}{2}$$

Since k is usually positive in an exponential decay model:

$$k = \frac{1}{\sqrt{2}}$$

ii. Marginal Probability Density Functions

The joint PDF is $f(x, y) = e^{-\frac{1}{\sqrt{2}}(x+y)}$ for $0 \leq y < x < \infty$.

- **Marginal PDF of X ($f_X(x)$):** Integrate $f(x, y)$ over all possible values of Y .
For X : $x \geq 0$. The range for Y is $0 \leq y < x$.

$$f_X(x) = \int_0^x e^{-k(x+y)} dy = e^{-kx} \int_0^x e^{-ky} dy$$

$$f_X(x) = e^{-kx} \left[\frac{e^{-ky}}{-k} \right]_0^x = e^{-kx} \left(\frac{e^{-kx} - 1}{-k} \right) = \frac{e^{-kx} - e^{-2kx}}{k}$$

Substituting $k = 1/\sqrt{2}$ (if $x > 0$, else $f_X(x) = 0$):

$$f_X(x) = \sqrt{2} \left(e^{-x/\sqrt{2}} - e^{-2x/\sqrt{2}} \right) \quad \text{for } x > 0$$

- **Marginal PDF of Y ($f_Y(y)$):** Integrate $f(x, y)$ over all possible values of X .
For Y : $y \geq 0$. The range for X is $y < x < \infty$.

$$f_Y(y) = \int_y^\infty e^{-k(x+y)} dx = e^{-ky} \int_y^\infty e^{-kx} dx$$

$$f_Y(y) = e^{-ky} \left[\frac{e^{-kx}}{-k} \right]_y^\infty = e^{-ky} \left(0 - \frac{e^{-ky}}{-k} \right) = \frac{e^{-2ky}}{k}$$

Substituting $k = 1/\sqrt{2}$ (if $y > 0$, else $f_Y(y) = 0$):

$$f_Y(y) = \sqrt{2} e^{-2y/\sqrt{2}} \quad \text{for } y > 0$$

iii. The Conditional Probability Density Functions of X and Y

The formula for conditional PDF is $f(x|y) = \frac{f(x,y)}{f_Y(y)}$ and $f(y|x) = \frac{f(x,y)}{f_X(x)}$.

- **$f(X|Y = y)$:** (Defined for $x > y$)

$$f(x|y) = \frac{e^{-k(x+y)}}{\frac{e^{-2ky}}{k}} = k \frac{e^{-kx} e^{-ky}}{e^{-2ky}} = k e^{-kx} e^{ky}$$

$$f(x|y) = \mathbf{k}e^{-k(x-y)} \quad \text{for } x > y$$

- $f(Y|X = x)$: (Defined for $0 < y < x$)

$$f(y|x) = \frac{e^{-k(x+y)}}{\frac{e^{-kx} - e^{-2kx}}{k}} = k \frac{e^{-kx} e^{-ky}}{e^{-kx}(1 - e^{-kx})}$$

$$f(y|x) = \frac{\mathbf{k}e^{-ky}}{\mathbf{1} - \mathbf{e}^{-kx}} \quad \text{for } 0 < y < x$$

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Great! Let's solve the remaining questions related to **Probability and Random Variables**. I'll include the **Weekly Assignment Questions** and the **Expected University Questions** as they cover the remaining content from your request.

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Weekly Assignment Questions

1. Probability Distribution of Doublets

Question: Find the probability distribution of the number of doublets in four throws of a pair of dice.

Solution:

1. **Define Success (p):** A doublet is when both dice show the same number:

$$\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}.$$

- Total outcomes when throwing a pair of dice: $6 \times 6 = 36$.
- Probability of success (p): $p = \frac{6}{36} = \frac{1}{6}$.
- Probability of failure (q): $q = 1 - p = 1 - \frac{1}{6} = \frac{5}{6}$.

2. **Binomial Parameters:** Number of trials $n = 4$.
3. **Random Variable (X):** Number of doublets in four throws. X can take values 0, 1, 2, 3, or 4.
4. **Calculate Probabilities using the Binomial formula:** $P(X = x) = \binom{n}{x} p^x q^{n-x}$.

X (Number of Doublets)	$P(X = x) = \binom{4}{x} \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{4-x}$	Probability (Fraction)
0	$\binom{4}{0} \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^4 = 1 \times 1 \times \frac{625}{1296}$	$\frac{625}{1296}$
1	$\binom{4}{1} \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^3 = 4 \times \frac{1}{6} \times \frac{125}{216}$	$\frac{500}{1296}$
2	$\binom{4}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^2 = 6 \times \frac{1}{36} \times \frac{25}{36}$	$\frac{150}{1296}$
3	$\binom{4}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^1 = 4 \times \frac{1}{216} \times \frac{5}{6}$	$\frac{20}{1296}$
4	$\binom{4}{4} \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^0 = 1 \times \frac{1}{1296} \times 1$	$\frac{1}{1296}$

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2. Finding Density Function from Distribution Function

Question: The probability distribution (CDF) of a random variable X is given by:

$$F(x) = \begin{cases} 1 - (1 + x)e^{-x}, & \text{for } x \geq 0 \\ 0, & \text{for } x < 0 \end{cases} \text{ Find the corresponding density function } (f(x)).$$

Solution:

The Probability Density Function (PDF) $f(x)$ is the derivative of the Cumulative Distribution Function (CDF) $F(x)$: $f(x) = \frac{d}{dx} F(x)$.

1. **Differentiate for $x \geq 0$:**

$$F(x) = 1 - (1 + x)e^{-x}$$

$$f(x) = \frac{d}{dx} [1 - (1 + x)e^{-x}]$$

$$f(x) = 0 - \left[\frac{d}{dx} (1 + x) \cdot e^{-x} + (1 + x) \cdot \frac{d}{dx} (e^{-x}) \right]$$

(Using the Product Rule: $(uv)' = u'v + uv'$)

$$f(x) = -[(1)e^{-x} + (1+x)(-e^{-x})]$$

$$f(x) = -[e^{-x} - e^{-x} - xe^{-x}]$$

$$f(x) = -[-xe^{-x}] = xe^{-x}$$

2. Define the PDF:

$$f(x) = \begin{cases} xe^{-x}, & \text{for } x \geq 0 \\ 0, & \text{for } x < 0 \end{cases}$$

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3. Finding Constant k and Probability Distribution (Cauchy)

Question: A random variable x has the density function $f(x) = \frac{k}{1+x^2}$, $-\infty < x < \infty$. Determine k and the prob. distribution.

Solution:

1. Determine k (using $\int_{-\infty}^{\infty} f(x)dx = 1$):

$$\int_{-\infty}^{\infty} \frac{k}{1+x^2} dx = 1$$

$$k[\arctan(x)]_{-\infty}^{\infty} = 1$$

$$k[\arctan(\infty) - \arctan(-\infty)] = 1$$

$$k\left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right] = 1$$

$$k(\pi) = 1 \implies k = \frac{1}{\pi}$$

2. Determine the Probability Distribution (CDF), $F(x)$:

$$F(x) = \int_{-\infty}^x f(t)dt = \int_{-\infty}^x \frac{1/\pi}{1+t^2} dt$$

$$F(x) = \frac{1}{\pi} [\arctan(t)]_{-\infty}^x$$

$$F(x) = \frac{1}{\pi} [\arctan(x) - \arctan(-\infty)]$$

$$F(x) = \frac{1}{\pi} \left[\arctan(x) - \left(-\frac{\pi}{2} \right) \right]$$

$$F(x) = \frac{1}{\pi} \arctan(x) + \frac{1}{2}$$

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4. Continuous Bivariate Joint PDF and Probability

Question: Suppose that two dimensional continuous random variable (X, Y) has joint pdf given by $f(x, y) = \begin{cases} 6x^2y, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$. (a) Verify that $\int_0^1 \int_0^1 f(x, y) dx dy = 1$. (b) Find $P(0 < X < \frac{3}{4}, \frac{1}{3} < Y < 2)$.

Solution:

(a) Verify total probability:

$$\int_0^1 \int_0^1 6x^2y dx dy = 1$$

Integrate with respect to x first:

$$\begin{aligned} &= \int_0^1 6y \left[\frac{x^3}{3} \right]_0^1 dy \\ &= \int_0^1 6y \left(\frac{1}{3} - 0 \right) dy = \int_0^1 2y dy \end{aligned}$$

Now integrate with respect to y :

$$= \left[2 \frac{y^2}{2} \right]_0^1 = [y^2]_0^1 = 1^2 - 0^2 = 1$$

The function is verified as a PDF.

(b) Find $P(0 < X < \frac{3}{4}, \frac{1}{3} < Y < 2)$:

The joint PDF is only defined for $0 < x < 1$ and $0 < y < 1$. Therefore, the upper limit for Y must be capped at 1.

$$P = \int_{1/3}^1 \left(\int_0^{3/4} 6x^2 y dx \right) dy$$

Integrate with respect to x :

$$\begin{aligned} &= \int_{1/3}^1 6y \left[\frac{x^3}{3} \right]_0^{3/4} dy \\ &= \int_{1/3}^1 2y \left[\left(\frac{3}{4} \right)^3 - 0 \right] dy = \int_{1/3}^1 2y \left(\frac{27}{64} \right) dy \\ &= \frac{54}{64} \int_{1/3}^1 y dy = \frac{27}{32} \left[\frac{y^2}{2} \right]_{1/3}^1 \end{aligned}$$

Now integrate with respect to y :

$$\begin{aligned} &= \frac{27}{32} \times \frac{1}{2} [y^2]_{1/3}^1 = \frac{27}{64} \left(1^2 - \left(\frac{1}{3} \right)^2 \right) \\ &= \frac{27}{64} \left(1 - \frac{1}{9} \right) = \frac{27}{64} \times \frac{8}{9} \\ &= \frac{3}{8} = \mathbf{0.375} \end{aligned}$$

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Expected University Questions

1. Possible Values for Difference in Heads and Tails

Question: Let X represents the difference between the number of heads and the number of tails obtained when a coin is tossed 6 times. What are possible values of X ?

Solution:

1. **Total Tosses:** $n = 6$.
2. Let H be the number of heads and T be the number of tails.
3. We know $H + T = 6$.
4. The random variable X is the difference: $X = |H - T|$.

We test the possible values for H :

H (Heads)	$T = 6 - H$ (Tails)	$X = H - T$ (Difference)
0	6	$0 - 6 = 6$
1	5	$1 - 5 = 4$
2	4	$2 - 4 = 2$
3	3	$3 - 3 = 0$
4	2	$4 - 2 = 2$
5	1	$5 - 1 = 4$
6	0	$6 - 0 = 6$

- **Possible values of X :** The possible values are **0, 2, 4, and 6**.



2. Finding Probability Distributions

Question: Find the probability distribution of:

(i) number of heads in two tosses of a coin

- (ii) number of tails in the simultaneous tosses of three coins
 (iii) number of heads in four tosses of a coin

Solution:

(i) Number of heads in two tosses ($n = 2, p = 1/2$):

X (Heads)	Outcomes	$P(X)$
0	TT	$1/4$
1	HT, TH	$2/4$
2	HH	$1/4$

(ii) Number of tails in the simultaneous tosses of three coins ($n = 3, p = 1/2$):

X (Tails)	Outcomes	$P(X) = \binom{3}{x}(1/2)^3$
0	HHH	$\binom{3}{0}/8 = 1/8$
1	HHT, HTH, THH	$\binom{3}{1}/8 = 3/8$
2	HTT, THT, TTH	$\binom{3}{2}/8 = 3/8$
3	TTT	$\binom{3}{3}/8 = 1/8$

(iii) Number of heads in four tosses of a coin ($n = 4, p = 1/2$):

X (Heads)	$P(X) = \binom{4}{x}(1/2)^4$
0	$\binom{4}{0}/16 = 1/16$
1	$\binom{4}{1}/16 = 4/16$
2	$\binom{4}{2}/16 = 6/16$
3	$\binom{4}{3}/16 = 4/16$
4	$\binom{4}{4}/16 = 1/16$

3. Probability Distribution for Die Success

Question: Find the probability distribution of the number of success in two tosses of a die, where a success is defined as (i) number greater than 4.

Solution:

1. **Define p and q :**

- Success (S): Getting a number greater than 4, i.e., $\{5, 6\}$.
- Probability of success (p): $p = \frac{2}{6} = \frac{1}{3}$.
- Probability of failure (q): $q = 1 - p = \frac{2}{3}$.

2. **Binomial Parameters:** Number of trials $n = 2$.

3. **Random Variable (X):** Number of successes in two tosses. X can take values 0, 1, or 2.

X (Number of Successes)	$P(X) = \binom{2}{x} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{2-x}$	Probability (Fraction)
0	$\binom{2}{0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^2 = 1 \times \frac{4}{9}$	$\frac{4}{9}$
1	$\binom{2}{1} \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^1 = 2 \times \frac{2}{9}$	$\frac{4}{9}$
2	$\binom{2}{2} \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^0 = 1 \times \frac{1}{9}$	$\frac{1}{9}$



4. PMF with Unknown Constant c

Question: A random variable X has the following probability mass function:

X	0	1	2
$p(x)$	$3c^3$	$4c - 10c^2$	$5c - 1$

I. Find c .

II. Evaluate $P(X < 2)$, $P(2X + 3 \geq 5)$ and $P(1 < X \leq 2)$.

III. If $P(X \leq a) < 1$, find the maximum value of a .

Solution:

I. Find c

Use the property $\sum P(x) = 1$:

$$3c^3 + (4c - 10c^2) + (5c - 1) = 1$$

$$3c^3 - 10c^2 + 9c - 2 = 0$$

By inspection, $c = 1$ is a root: $3(1) - 10(1) + 9(1) - 2 = 12 - 12 = 0$.

The factors are $(c - 1)(3c^2 - 7c + 2) = 0$.

$$(c - 1)(3c - 1)(c - 2) = 0$$

The roots are $c = 1, c = 1/3, c = 2$.

We must also ensure $P(x) \geq 0$.

- If $c = 2$, $P(X = 2) = 5(2) - 1 = 9 \geq 0$. $P(X = 1) = 4(2) - 10(4) = 8 - 40 = -32$, which is invalid.
- If $c = 1$, $P(X = 2) = 5(1) - 1 = 4$, $P(X = 1) = 4 - 10 = -6$, which is invalid.
- If $c = 1/3$:
 - $P(X = 0) = 3(1/3)^3 = 1/9 \geq 0$.
 - $P(X = 1) = 4(1/3) - 10(1/9) = 12/9 - 10/9 = 2/9 \geq 0$.
 - $P(X = 2) = 5(1/3) - 1 = 5/3 - 3/3 = 2/3 \geq 0$.
 - Check Sum: $1/9 + 2/9 + 6/9 = 9/9 = 1$. (Valid)
- **Therefore, $c = 1/3$.**

II. Evaluate Probabilities

- $P(X < 2)$: $P(X = 0) + P(X = 1)$

$$P(X < 2) = \frac{1}{9} + \frac{2}{9} = \frac{3}{9} \text{ or } \frac{1}{3}$$

- $P(2X + 3 \geq 5)$: This inequality is $2X \geq 2$, or $X \geq 1$.

$$P(X \geq 1) = P(X = 1) + P(X = 2) = \frac{2}{9} + \frac{6}{9} = \frac{8}{9}$$

- $P(1 < X \leq 2)$: This only includes $X = 2$.

$$P(1 < X \leq 2) = P(X = 2) = \frac{6}{9} \text{ or } \frac{2}{3}$$

III. If $P(X \leq a) < 1$, find the maximum value of a .

1. Find the cumulative probabilities:

- $P(X \leq 0) = 1/9$
- $P(X \leq 1) = 1/9 + 2/9 = 3/9$
- $P(X \leq 2) = 1/9 + 2/9 + 6/9 = 9/9 = 1$

2. We need $P(X \leq a) < 1$.

3. If $a = 2$, $P(X \leq 2) = 1$, which is **not** < 1 .

4. If $a = 1$, $P(X \leq 1) = 3/9 < 1$.

5. If a is any value in the interval $1 \leq a < 2$, $P(X \leq a) = P(X \leq 1) = 3/9 < 1$.

6. The smallest integer value of a that results in $P(X \leq a) = 1$ is $a = 2$. To keep the probability strictly less than 1, a must be less than 2.

- **The maximum integer value of a is 1.** (Since X is discrete, the function $P(X \leq a)$ jumps from $3/9$ to 1 at $a = 2$.)

✍

5. Joint Density, Marginal Density, and Independence

Question: If X and Y are two random variables having joint density function:

$$f(x, y) = \begin{cases} 6(1 - x - y) & 0 \leq x < 2, 2 \leq y < 4 \\ 0 & \text{elsewhere} \end{cases}$$

Find marginal density function of X and Y . Hence examine if X and Y are independent.

Solution:

The given limits $0 \leq x < 2$ and $2 \leq y < 4$ show the rectangular region of integration.

I. Find Marginal Density of X ($f_X(x)$):

$$f_X(x) = \int_2^4 f(x, y) dy \quad \text{for } 0 \leq x < 2$$

$$f_X(x) = \int_2^4 6(1 - x - y) dy = 6 \left[y - xy - \frac{y^2}{2} \right]_2^4$$

$$f_X(x) = 6 \left[\left(4 - 4x - \frac{16}{2} \right) - \left(2 - 2x - \frac{4}{2} \right) \right]$$

$$f_X(x) = 6 [(4 - 4x - 8) - (2 - 2x - 2)]$$

$$f_X(x) = 6 [(-4 - 4x) - (-2x)] = 6(-4 - 2x) = -24 - 12x$$

Wait! The marginal PDF must be ≥ 0 . Since $-24 - 12x$ is negative for $x \geq 0$, the original joint PDF definition must be wrong or transcribed incorrectly from the source material, as it violates the property that $\int f_X(x) dx = 1$. Assuming the question meant a different region or function, we proceed with the negative result to show the formal steps.

- $f_X(x) = -12x - 24$ for $0 \leq x < 2$.

II. Find Marginal Density of Y ($f_Y(y)$):

$$f_Y(y) = \int_0^2 f(x, y) dx \quad \text{for } 2 \leq y < 4$$

$$f_Y(y) = \int_0^2 6(1 - x - y) dx = 6 \left[x - \frac{x^2}{2} - xy \right]_0^2$$

$$f_Y(y) = 6 \left[\left(2 - \frac{4}{2} - 2y \right) - 0 \right]$$

$$f_Y(y) = 6(2 - 2 - 2y) = 6(-2y) = -12y$$

Again, the result is negative, indicating a mathematically invalid problem setup for a PDF.

- $f_Y(y) = -12y$ for $2 \leq y < 4$.

III. Examine Independence

X and Y are independent if $f(x, y) = f_X(x)f_Y(y)$.

$$f_X(x)f_Y(y) = (-12x - 24)(-12y) = 144y(x + 2)$$

$$f(x, y) = 6(1 - x - y)$$

Since $144y(x + 2) \neq 6(1 - x - y)$, the variables **are not independent** (even ignoring the negative probability issue). The mathematical dependence is obvious from the term $x - y$ in the original function.